Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, May - 2019
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) List the data transfer and data manipulation instructions available in a computer. [2]
b) What is an interrupt? What are the uses of interrupts? [3]
c) What is synchronous bus and asynchronous bus? [2]
d) What is hand shaking in I O organization? [3]
e) What are the characteristics of semiconductor RAM memories? [2]
f) A set associative cache consists of 128 blocks divided into 4 block/set. The main memory has 8192 blocks each consists of 128 words. How many address bits are required to access a main memory location? [3]
g) List various flags of 8086. [2]
h) Write the special functions of general purpose registers. [3]
i) List any four Unconditional Branch Instructions. [2]
j) What errors are present in the following: [3]

```
mov   ax, 3d
mov   23, ax
mov    cx, ch
move  ax, 1h
add    2, cx
add   3, 6
inc   ax, 2
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PART-B

(50 Marks)

2.a) Explain different functional units of a digital computer. Mention the functions of different processor registers i) IR ii) MAR iii) PC.
b) What is an addressing mode? List the different types of addressing modes. Register R5 is used in a program to point to the top of a stack. Write a sequence of instructions using the Index, Autoincrement, and Autodecrement addressing modes to perform each of the following tasks:
i) Pop the top two items of the stack, add them, and then push the result onto the stack.
ii) Copy the fifth item from the top into register R3.
iii) Remove the top ten items from the stack. [5+5]

OR

3.a) Explain the different instruction formats.

b) Mention four types of operations required to perform by instructions in a computer. Show how the operation C = A + B can be implemented using:
i) Three address instruction
ii) Two address instruction
iii) One address instruction. [5+5]

4.a) What is the difference between isolated I/O and memory-mapped I/O? What are the advantages and disadvantages of each?

b) How many characters per second can be transmitted over a 1200-baud line in each of the following modes (Assume a character code of eight bits).
i) Synchronous serial transmission
ii) Asynchronous serial transmission with two stop bits
iii) Asynchronous serial transmission with one stop bit. [5+5]

OR

5. Explain with a block diagram the DMA transfer in a computer system. [10]

6.a) Explain in detail about the classification of memory.

b) Given the hit ratio of 0.92 and cache access time of 40ns and main memory access time of 300 ns. Calculate the average access time. Discuss how read and write operations are carried out in a cache memory? [5+5]

OR

7. Explain in detail about different memory mapping techniques. [10]

8.a) Explain the principles and characteristics of Pipelining.

b) What are the various segments registers in 8086 microprocessor? [7+3]

OR

9.a) Draw the functional pin diagram of 8086. Briefly explain the action of 8086 when NMI and INTR pins are activated.

b) Register R1 and R2 of computer contain the decimal value 1200 and 4600. What is the effective address of the source operand in each of the following instructions?
i) Load 20(R1), R5
ii) Move #3000, R5
iii) Store R5, 30(R1, R2)
iv) Add –(R2), R5
v) Subtract (R1)+, R5. [5+5]

10. Write an ALP in 8086 to sort a given set of 8-bit unsigned integers into ascending order by bubble sort method. [10]

OR

11.a) Explain the following 8086 instructions with examples.
i) RET ii) LOOP iii) IRET.

b) What is importance of CALL Instruction?

c) Write assembly language program to list alphanumeric characters by illustrating JUMP instruction? [3+2+5]

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